

2.4 行列式按某一行(列)展开

2025年8月25日 23:55

例3. 注意到每一行都有 $x_1, x_2, \dots, x_n$ 加边法.

$$\begin{vmatrix} x_1 - a_1 & x_2 & \cdots & x_n \\ x_1 & x_2 - a_2 & \cdots & x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_1 & x_2 & \cdots & x_n - a_n \end{vmatrix}$$

$$|A| = \begin{vmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ 0 & x_1 - a_1 & x_2 & \cdots & x_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & x_1 & x_2 & \cdots & x_n - a_n \end{vmatrix} = \begin{vmatrix} 1 & x_1 & x_2 & \cdots & x_n \\ -1 & -a_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ -1 & \cdots & \cdots & -a_n \end{vmatrix}$$

$$= \begin{vmatrix} 1 - \frac{x_1}{a_1} - \frac{x_2}{a_2} - \cdots - \frac{x_n}{a_n} & x_1 & x_2 & \cdots & x_n \\ 0 & -a_1 & \cdots & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & -a_n \end{vmatrix} = (-1)^n \left(1 - \sum_{i=1}^n \frac{x_i}{a_i} \right) a_1 a_2 \cdots a_n$$

例4 计算 n 阶行列式 ($n \geq 2$):

$$D_n = \begin{vmatrix} x & 0 & 0 & \cdots & 0 & 0 & a_0 \\ -1 & x & 0 & \cdots & 0 & 0 & a_1 \\ 0 & -1 & x & \cdots & 0 & 0 & a_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & x & a_{n-2} \\ 0 & 0 & 0 & \cdots & 0 & -1 & x + a_{n-1} \end{vmatrix}$$

书本上更严谨的证明:

采用数学归纳法.

解 $n=2$ 时,

$$D_2 = \begin{vmatrix} x & a_0 \\ -1 & x + a_1 \end{vmatrix} = x^2 + a_1 x + a_0$$

假设对于上述形式的 $n-1$ 阶行列式, 有

$$\begin{vmatrix} x & 0 & \cdots & 0 & 0 & a_0 \\ -1 & x & \cdots & 0 & 0 & a_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & -1 & x + a_{n-2} \end{vmatrix} = x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0$$

现在来看上述形式的 n 阶行列式, 把它按第 1 行展开, 得

$$\begin{aligned}
 D_n &= x \begin{vmatrix} -1 & x & 0 & \cdots & 0 & 0 \\ 0 & -1 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & x \end{vmatrix} + (-1)^{1+n} a_0 \begin{vmatrix} -1 & x & 0 & \cdots & 0 & 0 \\ 0 & -1 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & x \end{vmatrix} \\
 &= x(x^{n-1} + a_{n-1}x^{n-2} + \cdots + a_2x + a_1) + (-1)^{1+n} a_0 (-1)^{n-1} \\
 &= x^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0
 \end{aligned}$$

$$D_n = (x + a_{n-1}) \begin{vmatrix} x & & & & \\ -1 & x & & & \\ & -1 & x & & \\ & & \ddots & \ddots & \\ & & & -1 & x \end{vmatrix} + \begin{vmatrix} x & & & & a_0 \\ -1 & x & & & a_1 \\ & -1 & x & & a_2 \\ & & \ddots & \ddots & \vdots \\ & & & -1 & x + a_{n-2} \end{vmatrix}$$

$$= x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0$$

例5 计算 n 阶行列式:

$$D_n = \begin{vmatrix} 2 & -1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & \cdots & 0 & -1 & 2 \end{vmatrix}$$

$$D_n = \begin{vmatrix} 1 & -1 & 0 & \cdots & 0 \\ 0 & 2 & -1 & \cdots & 0 \\ 0 & -1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 2 \end{vmatrix} = D_{n-1} + 1 \cdot (-1)^{1+n} \cdot (-1)^{n-1} = D_{n-1} + 1$$

$$D_1 = 2 \quad \therefore D_n = n+1$$

例题部分都非常重要!!!

习题部分:

2. 计算 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} a_1 & a_2 & a_3 & \cdots & a_{n-1} & a_n \\ 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 2 & -2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & n-1 & 1-n \end{vmatrix}$$

$$|A| = \begin{vmatrix} \sum a_i & a_2 & a_3 & \cdots & a_{n-1} & a_n \\ 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 2 & -2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & n-1 & 1-n \end{vmatrix} = \sum a_i \begin{vmatrix} -1 & 0 & \cdots & 0 & 0 \\ 2 & -2 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & n-1 & 1-n \end{vmatrix} = \sum a_i \begin{vmatrix} -1 & \cdots & \cdots & 0 & 0 \\ 0 & -2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & n-1 & 1-n \end{vmatrix}$$

$$= \cdots = \left(\sum a_i \right) (-1)^{n-1} (n-1)!$$

3. 计算 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} \end{vmatrix}.$$

Vandermonde 行列式:

$$= \prod_{1 \leq i < j \leq n} (a_j - a_i)$$

4. 计算 n 阶行列式:

$$D_n = \begin{vmatrix} 2a & a^2 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & 2a & a^2 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 2a & a^2 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 2a & a^2 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 2a \end{vmatrix}.$$

$$D_n = 2a D_{n-1} + (-1) \begin{vmatrix} a^2 & 0 & 0 & \cdots & 0 \\ 1 & 2a & a^2 & \cdots & 0 \\ & 1 & 2a & \cdots & a^2 \\ & & \ddots & \ddots & 2a \end{vmatrix} = 2a D_{n-1} - a^2 D_{n-2}$$

$$\Rightarrow D_n - a D_{n-1} = a(D_{n-1} - a D_{n-2}).$$

$$\therefore D_2 - a D_1 = 3a^2 - 2a^2 = a^2 \quad \therefore \text{有方程组} \quad \begin{cases} D_n - a D_{n-1} = a^n \\ D_{n-1} - a D_{n-2} = a^{n-1} \\ \vdots \\ D_2 - a D_1 = a^2 \end{cases} \rightarrow \begin{cases} D_n - a D_{n-1} = a^n \\ a D_{n-1} - a^2 D_{n-2} = a^n \\ \vdots \\ a^{n-2} D_2 - a^{n-1} D_1 = a^n \end{cases}$$

左式全部相加, 得 $D_n - a^{n-1}D_1 = (n+1)a^n$, 即 $D_n = (n+1)a^n$

5. 解方程:

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ x & a_1 & \cdots & a_{n-1} \\ x^2 & a_1^2 & \cdots & a_{n-1}^2 \\ \vdots & \vdots & & \vdots \\ x^{n-1} & a_1^{n-1} & \cdots & a_{n-1}^{n-1} \end{vmatrix} = 0,$$

$x = a_1$ 或 a_2 或 $\cdots$ 或 a_{n-1}

其中 $a_1, a_2, \cdots, a_{n-1}$ 是两两不等的数。

6. 计算 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & 2 & 2 & \cdots & 2 & 2 & 2 \\ 2 & 2 & 2 & \cdots & 2 & 2 & 2 \\ 2 & 2 & 3 & \cdots & 2 & 2 & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 2 & 2 & 2 & \cdots & 2 & n-1 & 2 \\ 2 & 2 & 2 & \cdots & 2 & 2 & n \end{vmatrix}$$

$$|A| = \begin{vmatrix} -1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 2 & 2 & 2 & \cdots & 2 & 2 & 2 \\ 0 & 0 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & n-3 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & n-2 \end{vmatrix} = (-1) \cdot 2 \cdot (n-2)! = -2(n-2)!$$

7. 计算 n 阶行列式:

$$D_n = \begin{vmatrix} x & y & y & \cdots & y & y \\ z & x & y & \cdots & y & y \\ z & z & x & \cdots & y & y \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ z & z & z & \cdots & x & y \\ z & z & z & \cdots & z & x \end{vmatrix}, y \neq z.$$

每一行加下一行的 (-1) 倍。

$$D_n = \begin{vmatrix} x-z & y-x & 0 & \cdots & 0 & 0 \\ 0 & x-z & y-x & \cdots & 0 & 0 \\ 0 & 0 & x-z & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & x-z & y-x \\ z & z & z & \cdots & z & x \end{vmatrix}$$

$$\therefore D_n = (x-z)D_{n-1} + (-1)^{n+1}z(y-x)^{n-1}.$$

$$= (x-z)D_{n-1} + z(x-y)^{n-1} \quad \textcircled{1} \quad y \text{ 与 } z \text{ 地位相同. 或说, } |A| = |A'|, \text{ 从而有}$$

$$D_n = (x-y)D_{n-1} + y(x-z)^{n-1} \quad \textcircled{2}$$

$$\text{联立 } \textcircled{1} \textcircled{2} \text{ 式, 得到 } D_n = \frac{y(x-z)^n - z(x-y)^n}{y-z}$$

另一个比较难想的方法, 来自课后考考答案。

解: 记原 n 阶行列式为 D_n , 将行列式按最后一列写成两个行列式的和

$$= \begin{vmatrix} x & y & y & \cdots & y & y & y \\ z & x & y & \cdots & y & y & y \\ z & z & x & \cdots & y & y & y \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ z & z & z & \cdots & z & x & y \\ z & z & z & \cdots & z & z & y \end{vmatrix} + \begin{vmatrix} x & y & y & \cdots & y & y & 0 \\ z & x & y & \cdots & y & y & 0 \\ z & z & x & \cdots & y & y & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ z & z & z & \cdots & z & x & 0 \\ z & z & z & \cdots & z & z & x-y \end{vmatrix}$$

$$= \begin{vmatrix} x & y & y & \cdots & y & y & y \\ z-x & x-y & 0 & \cdots & 0 & 0 & 0 \\ 0 & z-x & x-y & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & z-x & x-y & 0 \\ 0 & 0 & 0 & \cdots & 0 & z-x & 0 \end{vmatrix} + (x-y) \begin{vmatrix} x & y & y & \cdots & y & y \\ z & x & y & \cdots & y & y \\ z & z & x & \cdots & y & y \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ z & z & z & \cdots & z & x \\ z & z & z & \cdots & z & z \end{vmatrix}_{n-1}$$

$$= (-1)^{n+1}y(z-x)^{n-1} + (x-y)D_{n-1}$$

即: $D_n = (x-y)D_{n-1} + (-1)^{n+1}y(z-x)^{n-1}$

8. 计算 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 2 & 3 & 4 & \cdots & n & 1 \\ 3 & 4 & 5 & \cdots & 1 & 2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ n & 1 & 2 & \cdots & n-2 & n-1 \end{vmatrix}$$

与例7非常相似。

$$|A| = \begin{vmatrix} 1 & 2 & 3 & \cdots & n-1 & n \\ 1 & 1 & 1 & \cdots & 1 & 1-n \\ 1 & 1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1-n & 1 & \cdots & 1 & 1 \end{vmatrix} = \begin{vmatrix} \frac{n(n+1)}{2} & 2 & 3 & \cdots & n-1 & n \\ 0 & 1 & 1 & \cdots & 1 & 1-n \\ 0 & 1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 1-n & 1 & \cdots & 1 & 1 \end{vmatrix}$$

$$= \frac{n(n+1)}{2} \begin{vmatrix} 1 & 1 & \cdots & 1 & 1-n \\ 1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1-n & 1 & \cdots & 1 & 1 \end{vmatrix} = \frac{n(n+1)}{2} \begin{vmatrix} -1 & 1 & \cdots & 1 & 1-n \\ -1 & 1 & \cdots & 1-n & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & 1 & \cdots & 1 & 1 \end{vmatrix} = \frac{n(n+1)}{2} (-1) \begin{vmatrix} 1 & 0 & \cdots & 0 & -n \\ 1 & 0 & \cdots & -n & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & -n & \cdots & 1 & 1 \\ 1 & 0 & \cdots & 0 & 0 \end{vmatrix}_{n-1}$$

$$= \frac{n(n+1)}{2} (-1) \cdot (-n)^{n-2} \cdot (-1)^{(n-1, n-2, \dots, 1)} = \frac{n(n+1)}{2} (-1)^{\frac{(n-2)(n-1)}{2} + (n-1)} n^{n-2} = \frac{n^{n-1}(n+1)}{2} \cdot (-1)^{\frac{n(n-1)}{2}}$$

9. 用本节典型例题的例8的结果, 计算下列 n 阶行列式:

$$(1) \begin{vmatrix} 1+x_1y_1 & 1+x_1y_2 & \cdots & 1+x_1y_n \\ 1+x_2y_1 & 1+x_2y_2 & \cdots & 1+x_2y_n \\ \vdots & \vdots & \ddots & \vdots \\ 1+x_ny_1 & 1+x_ny_2 & \cdots & 1+x_ny_n \end{vmatrix};$$

$$(2) \begin{vmatrix} 1+t & t & t & \cdots & t \\ t & 2+t & t & \cdots & t \\ t & t & 3+t & \cdots & t \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ t & t & t & \cdots & n+t \end{vmatrix}.$$

例8结果

例8 设数域 K 上 n 级矩阵 $A = (a_{ij})$, 它的 (i, j) 元的代数余子式记作 A_{ij} . 把 A 的每个元素都加上同一个数 t , 得到的矩阵记作 $A(t) = (a_{ij} + t)$. 证明:

$$|A(t)| = |A| + t \sum_{i=1}^n \sum_{j=1}^n A_{ij}.$$

其证明:

$$|A(t)| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} t & a_{12} & \cdots & a_{1n} \\ t & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ t & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$+ \cdots + \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1, n-1} & t \\ a_{21} & a_{22} & \cdots & a_{2, n-1} & t \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{n, n-1} & t \end{vmatrix}$$

$$9. (1) \quad n=1 \quad |A(1)| = 1+x_1y_1$$

$$n=2 \quad |A(1)| = (1+x_1y_1)(1+x_2y_2) - (1+x_1y_2)(1+x_2y_1)$$

$$n=2 \quad |A(t)| = (1+x_1 y_1)(1+x_2 y_2) - (1+x_1 y_2)(1+x_2 y_1) \\ = (x_1 - x_2)(y_1 - y_2)$$

$$n \geq 3 \text{ 时, } |A| = 0 \quad A_{kl} = 0 \quad \therefore |A(t)| = 0$$

$$(2) \quad |A| = n! \quad A_{kk} = \frac{n!}{k} \quad \text{从而 } |A(t)| = n! + t \left(n! + \frac{n!}{2} + \frac{n!}{3} + \cdots + \frac{n!}{n} \right) = n! + n! t \sum_{k=1}^n \frac{1}{k}$$

$$+ \cdots + \begin{vmatrix} a_{21} & a_{22} & \cdots & a_{2,n-1} & t \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{n,n-1} & t \end{vmatrix} \\ = |A| + tA_{11} + tA_{21} + \cdots + tA_{n1} + \cdots + tA_{1n} + tA_{2n} + \cdots + tA_{nn} \\ = |A| + t \sum_{i=1}^n \sum_{j=1}^n A_{ij}.$$

10. 计算 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & a_1 & 0 & \cdots & 0 \\ 1 & 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & \cdots & a_{n-1} \end{vmatrix},$$

其中 $a_1 a_2 \cdots a_{n-1} \neq 0$.

$$|A| = \begin{vmatrix} 1 - \frac{1}{a_1} & \cdots & -\frac{1}{a_{n-1}} & 1 & \cdots & 1 \\ 0 & & a_1 & & & \\ 0 & & & a_2 & & \\ \vdots & & & & \ddots & \\ 0 & & & & & a_{n-1} \end{vmatrix} = \left(1 - \sum_{i=1}^{n-1} \frac{1}{a_i} \right) a_1 a_2 \cdots a_{n-1}$$

11. 计算 n 阶行列式:

$$D_n = \begin{vmatrix} 5 & 3 & 0 & 0 & \cdots & 0 & 0 \\ 2 & 5 & 3 & 0 & \cdots & 0 & 0 \\ 0 & 2 & 5 & 3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 & 5 \end{vmatrix}$$

$$D_n = 5D_{n-1} - 3 \begin{vmatrix} 2 & 3 \\ 0 & 5 & 3 \\ \vdots & \ddots & \vdots \\ 0 & & 5 \end{vmatrix} = 5D_{n-1} - 6D_{n-2} \quad \text{得} \begin{cases} D_n - 3D_{n-1} = 2(D_{n-1} - 3D_{n-2}) \\ D_n - 2D_{n-1} = 3(D_{n-1} - 2D_{n-2}) \end{cases}$$

$$D_1 = 5 \quad D_2 = 19 \quad \begin{cases} D_2 - 3D_1 = 4 \\ D_2 - 2D_1 = 9 \end{cases} \quad \text{得} \begin{cases} D_n - 3D_{n-1} = 4 \cdot 2^{n-2} = 2^n \\ D_n - 2D_{n-1} = 9 \cdot 3^{n-2} = 3^n \end{cases}$$

$$\text{解得 } D_n = 3^{n+1} - 2^{n+1}$$

12. 计算 n 阶行列式:

$$D_n = \begin{vmatrix} 1+x^2 & x & 0 & 0 & \cdots & 0 & 0 \\ x & 1+x^2 & x & 0 & \cdots & 0 & 0 \\ 0 & x & 1+x^2 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & x & 1+x^2 \end{vmatrix}.$$

$$D_n = (1+x^2)D_{n-1} - x \begin{vmatrix} x & & & \\ x & 1+x^2 & & \\ & & 1+x^2 & \\ & & & 1+x^2 \end{vmatrix} = (1+x^2)D_{n-1} - x^2 D_{n-2}.$$

$$D_2 = (1+x^2)^2 - x^2 = x^4 + x^2 + 1.$$

$$D_1 = 1+x^2 \quad D_2 - x^2 D_1 = 1.$$

得 $\begin{cases} D_n - D_{n-1} = x^2 (D_{n-1} - D_{n-2}) \\ D_n - x^2 D_{n-1} = 1 \end{cases} \Rightarrow \begin{cases} D_n - D_{n-1} = x^4 \cdot x^{2(n-2)} = x^{2n} \\ D_n - x^2 D_{n-1} = 1 \end{cases}$

解得 $D_n = \frac{x^{2(n+1)} - 1}{x^2 - 1} = \frac{(1+x^2+x^4+\dots+x^{2n})(x^2-1)}{x^2-1} = 1+x^2+x^4+\dots+x^{2n}$

13. 计算 n 阶行列式 ($n \geq 2$):

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1+1 & x_2+1 & \cdots & x_n+1 \\ x_1^2+x_1 & x_2^2+x_2 & \cdots & x_n^2+x_n \\ x_1^3+x_1^2 & x_2^3+x_2^2 & \cdots & x_n^3+x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1}+x_1^{n-2} & x_2^{n-1}+x_2^{n-2} & \cdots & x_n^{n-1}+x_n^{n-2} \end{vmatrix}.$$

$$|A| = \begin{vmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{vmatrix} = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

* 14. 计算 n 阶行列式:

$$\begin{vmatrix} 1-a_1 & a_2 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1-a_2 & a_3 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_3 & a_4 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1-a_n \end{vmatrix}.$$

$$D_n = (1-a_n)D_{n-1} + (-1)^{2n-1} \begin{vmatrix} 1-a_1 & a_2 & & & \\ -1 & 1-a_2 & a_3 & & \\ & -1 & 1-a_3 & a_4 & \\ & & & \ddots & \vdots \\ & & & & -1 & 1-a_n \end{vmatrix} = (1-a_n)D_{n-1} + a_n D_{n-2}.$$

从而有 $\begin{cases} D_n - D_{n-1} = -a_n (D_{n-1} - D_{n-2}) \\ D_{n-1} - D_{n-2} = -a_{n-1} (D_{n-2} - D_{n-3}) \\ \vdots \\ D_3 - D_2 = -a_3 (D_2 - D_1) \end{cases}$ 得 $D_n - D_{n-1} = (-1)^{n-2} a_1 a_2 \cdots a_n$
 $= (-1)^n a_1 a_2 \cdots a_n$

$$-(-1)^n a_1 a_2 \cdots a_n$$

$$\begin{cases} D_3 - D_2 = -a_3 (D_2 - D_1) \\ D_2 - D_1 = a_1 a_2 \end{cases}$$

从而有

$$\begin{cases} D_2 - D_1 = a_1 a_2 \\ D_3 - D_2 = -a_1 a_2 a_3 \\ D_4 - D_3 = a_1 a_2 a_3 a_4 \\ \vdots \\ D_n - D_{n-1} = -a_1 a_2 a_3 \cdots a_n \end{cases} \quad \begin{aligned} \therefore D_n - D_1 &= a_1 a_2 - a_1 a_2 a_3 + a_1 a_2 a_3 a_4 - \cdots + (-1)^n a_1 a_2 \cdots a_n \\ \therefore D_n &= a_1 a_2 - a_1 a_2 a_3 + \cdots + (-1)^n a_1 a_2 \cdots a_n + 1 - a_1 \end{aligned}$$